MILITARY STANDARD

LUBRICATION INSTRUCTIONS, PREPARATION AND PRESENTATION OF

UNITED STATES
GOVERNMENT PRINTING OFFICE
WASHINGTON: 1960

For sale by the Superintendent of Documents, U.S. Government Printing Office
Washington 25, D. C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

ARMED FORCES SUPPLY SUPPORT CENTER

WASHINGTON 25, D. C.

Lubrication Instructions, Preparation and Presentation Of
MIL–STD–33

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 2 November 1960.

2. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Armed Forces Supply Support Center, Washington 25, D. C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

CONTENTS

		Page
1	SCOPE	1
1.1	Scope	1
1.2	Application	1
2	REFERENCED DOCUMENTS	2
3	DEFINITIONS	3
3.1	Definitions	3
3.2	Lubrication section	3
3.3	Lubrication order	3
3.3.1	Lubrication manual	3
3.3.2	Lubrication chart	3
3.3.3	Decalcomania	3
3.3.4	Plates	3
4	GENERAL REQUIREMENTS	4
4.1	Title	4
4.1.1	Identification	4
4.2	Security classification	4
4.3	Printing	4
4.4	Lubricants	4
4.5	Interval	4
4.5.1	Intervals and symbols	4
4.5.1.1	Combined interval	4
5	DETAILED REQUIREMENTS	5
5.1	Lubrication section	5
5.2	Lubrication manual	5
5.3	Instructions	5
5.3.1	Presentation	5
5.4	Lubrication charts	5
5.4.1	Size	5
5.4.2	Lamination	5
5.4.3	Content	5
5.4.3.1	Supplemental information	6
5.5	Decalcomania	6
5.6	Plates	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers the general and detailed requirements for the preparation of lubrication instructions. These instructions cover the lubrication and other servicing points, lubricants, and procedures for periodic lubrication of all military equipment.

1.2 Application. The requirements of this standard shall apply to lubrication instructions prepared by or for military departments or services. The published instructions may be designated as lubrication order, lubrication chart, lubrication diagram, or by any other title deemed appropriate by the responsible department or service.

1.2.1. This standard shall apply to mandatory lubrication instructions for all equipment which requires periodic lubrication. These instructions are to be limited to periodic lubrication services only. Instructions that are a part of preservation in storage or of normal maintenance operations (such as coating parts with oil prior to installation) will be included in the appropriate section of the applicable maintenance manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

2. REFERENCE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this standard.

SPECIFICATION

MILITARY
MIL–F–1040 —Film, Flexible. Vinyl.

STANDARDS

MILITARY
MIL–STD–218–2—Production or Procurement of Artwork for Technical Manuals.
MIL–STD–218–3—Preparation of Manuscript (Final Typed) for Technical Manuals.
MIL–STD–290 —Packaging, Packing and Marking of Petroleum and Related Products.

(Copies of specifications or standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply:

Federal Supply Catalog C4–1 —Petroleum Base Products and Related Materials.

(Application for copies should be addressed to: Executive Director, Military Petroleum Supply Agency, Washington 25, D. C., Attention Code 73.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Definitions. The following definitions are intended to provide more accurate means of reference within, and for use of this standard, but are not intended as mandatory titles.

3.2 Lubrication section. That section or chapter of the appropriate manual covering the complete lubrication instructions for the end item or component.

3.3 Lubrication order. A separately published, concise document containing condensed lubrication instruments and prescribing lubricants and lubrication intervals, and intended to be attached to or carried with the equipment.

3.3.1 *Lubrication Manual.* A separate manual covering the lubrication of the end item in detail and should be used to provide lubrication instructions for large or complex equipment.

3.3.2 *Lubrication chart.* A card-type instruction prescribing all lubrication intervals, lubricants, and the necessary instructions for their application. A lubrication chart may consist of one or more cards and should be used for items that do not require detailed instructions.

3.3.3 *Decalcomania.* A lubrication chart in the form of a decalcomania. Lubrication instructions may be presented in the form of a decalcomania for items that are of such size or that by intended use render the other forms of instructions impractical. An example of this item would be a portable generator. Decalcomanias may also be used to supplement the other forms of lubrication instructions.

3.3.4 *Plates.* Plastic or metal plates containing specific lubrication instructions permanently fastened to the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Title. The title of all lubrication instructions shall contain the approved Federal nomenclature and military model number in the same form as the maintenance manual for the item.

4.1.1 *Identification.* Lubrication orders shall be dated. Further identification such as by manufacturers name and model number shall be as specified by the responsible department or service.

4.2 Security Classification. Lubrication instructions classified CONFIDENTIAL or higher shall not be published in any manual unless the manual has the same classification. When the lubrication instructions have a different classification than the manual, a separate media shall be used and given the proper classification.

4.3 Printing. All lubrication instructions shall be prepared for direct photographic reproduction or for photo-offset type printing. Size and style of type will be selected to provide the user with clear, legible instructions. Normally, all type, when reproduced, will be 8-point with a minimum of 6-point. Only medium and boldface types will be used. Preparation of artwork and manuscript shall be in accordance with MIL–STD–218–2 and MIL–STD–218–3, respectively.

4.4 Lubricants. All servicing materials specified shall be approved for military use and shall be identified by the approved symbol and military specification in accordance with Federal Supply Catalog C4–1 and the appendix to Military Standard MIL–STD–290. Each symbol shall be explained the first time that it appears in narrative form and in the key in illustrations or card type orders. Example for narrative: "Wheel bearings shall be packed with GAA (Grease, Automotive and Artillery) MIL–G–10924." For example of Key see figure 2a.

4.5 Interval. A definite interval shall be clearly specified for each lubrication point. Lubrication intervals, within each lubrication instruction, should correlate with the end item preventive maintenance and inspection intervals insofar as practical. To this end, consideration shall be given to reducing, rather than extending the interval. Standardization of intervals for each end item of equipment and between related end items will simplify scheduling by using organizations and it will assure a greater degree of compliance with the instructions.

4.5.1 *Intervals and symbols.* The following intervals and their approved symbol for use in illustrations shall be used. Only those intervals as are necessary shall be specified. When the responsible department or service has a requirement for an interval that is not shown, an appropriate interval and symbol may be developed. An explanation of all symbols that appear on each illustration shall be given in the Key (fig. 2a).

Calendar Interval	*Symbol*	*Miscellaneous Interval*	*Symbol*
Daily	D	Before operating	B.O.
Weekly	W	Before firing	B.F.
Monthly	M	After firing	A.F.
Quarterly (3 months)	Q	At assembly	A.A.
Semiannually (6 months)	S	At overhaul	A.O.
Annually (12 months)	A		

Mileage Interval	*Symbol*	*Operational Hours Interval*	*Symbol*
250	¼	4	4H
500	½	8	8H
750	¾	30	30H
1000	1	50	50H
2000	2	120	120H
3000	3	200	200H
6000	6	360	360H
12000	12	500	500H
		720	720H
		1000	1000H

4.5.1.1 *Combined interval.* Intervals may be combined when the lubrication requirements of the item indicate that a single interval is inadequate. Example: "3 months, 2000 miles, or 50 hours whichever occurs first." Combined intervals shall be avoided wherever possible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 Lubrication section. Unless otherwise specified by the responsible department or service, a lubrication section shall be included in the maintenance manual for the item. The information in this section shall be complete and specific. The format shall be the same as for the lubrication manual (5.2) except that the lubrication section shall be expanded to include a reproduction of any other form of lubrication order such as charts, plates or decalcomania pertaining to the item. When the lubrication order is produced in the form of a separate manual, the lubrication order will not be reproduced in the maintenance manual unless it is so specified by the responsible department or service.

5.2 Lubrication manual. Lubrication manuals shall contain complete instructions for lubrication of the item. This manual shall be the same size as specified for the corresponding maintenance manual. Lubrication instructions shall be published as a manual when the equipment is complex and would require a great number of pages of instructions. One or more illustrations shall be prepared for this manual to clearly locate the lubrication points on the equipment with close up views when required to assure clarity (fig. 1). Complete instructions on each point will be provided in the illustration or on the opposite page to eliminate the need for the user to refer to several pages. Illustrations will not be required for lubrication points that are obvious, such as the location of a wheel bearing; however, the proper method of servicing these points must be specified.

5.3 Instructions. Instructions shall include the following:

(1) Adjusting intervals for unusual conditions (dusty areas, etc.)

(2) Changing of lubricants for temperature variations.

(3) Cleaning fittings prior to lubrication.

(4) Specific instructions on lubrication services, i.e., how to pack wheel bearings.

(5) How to report unsatisfactory equipment.

(6) Any lubrication requirement that is peculiar to the item.

(7) Caution notes for points that must not be overlubricated or that appear to require lubrication but are not actually a lubrication point.

(8) The type, grade (and quantity where applicable) of lubricant.

(9) The lubrication interval.

((1) and (9) may be given in the form shown in KEY (fig. 2)).

5.3.1 *Presentation.* Instructions may be presented in narrative form with reference to lubrication points keyed to localized views of the point, by means of a series of illustrations, or a combination of these forms. The description of each point and the total number of identical points as a parenthetical note shall be given. Example for narrative: "Lubricate the rear spring rear shackles (2 each side) with GAA" Example for illustration: "Spring shackle R (2 ea.)"

Lubrication points shall be grouped by location and logical sequence. Example: All points under the hood, in the turret, under the front of the vehicle, inside of a compartment, etc.

5.4 Lubrication Charts. Lubrication charts may be prepared for items that can be adequately covered by this form of order. One or more cards printed both sides may be used.

5.4.1 *Size.* Chart size shall be as specified by the responsible department or service. The chart should be large enough to provide instructions and views that are clear and uncrowded. The use of two or more easily understood charts is preferred to one cluttered chart. When specifying chart size, the legibility of the reduced prints that may be required for the maintenance manual should be considered.

5.4.2 *Lamination.* When specified, lubrication charts shall be laminated. Laminating material shall conform to type II, class 1 of MIL–F–10400, two to six sheets thick.

5.4.3 *Content.* Lubrication charts shall contain all the information required for lubrication manuals except that they shall consist of illus-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

trations only, with explanatory notes (figs. 2 and 2a).

5.4.3.1 *Supplemental information.* When an item contains one or more lubrication points or a self-contained component that would be difficult to include in the chart, supplemental instructions may be presented in the form of a decalcomania. When decalcomania are used for this purpose, their location on the item shall be specified. This location shall be accessible, and it shall be protected. Examples are: under the hood, inside of a compartment door, on the back of an access panel, etc. The location shall be indicated on the chart by a note with a leader line indicating the location on the item.

5.5 Decalcomania. Decalcomania shall be prepared in the same manner and format as lubrication charts. Their application shall be limited to small or simple items that will not require extensive instructions and therefore the use of close up views may be eliminated. When used as supplemental information, notation shall be made that the decalcomania is a supplement and the document that it supplements shall be referred to.

5.6 Plates. Plastic or metal plates which are supplied by the manufacturer may be allowed to remain on the item provided that military lubricants, and the required changes for temperature variation are prescribed. Plastic or metal plates shall not be prepared or procured by military departments or services for installation on an item except for replacement purposes.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Copies of this standard for military use may be obtained as indicated in the foreword to, or the general provisions of the Index of Military Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ord
Navy—Wep
Air Force—AMC

Preparing Activity:
Army—Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

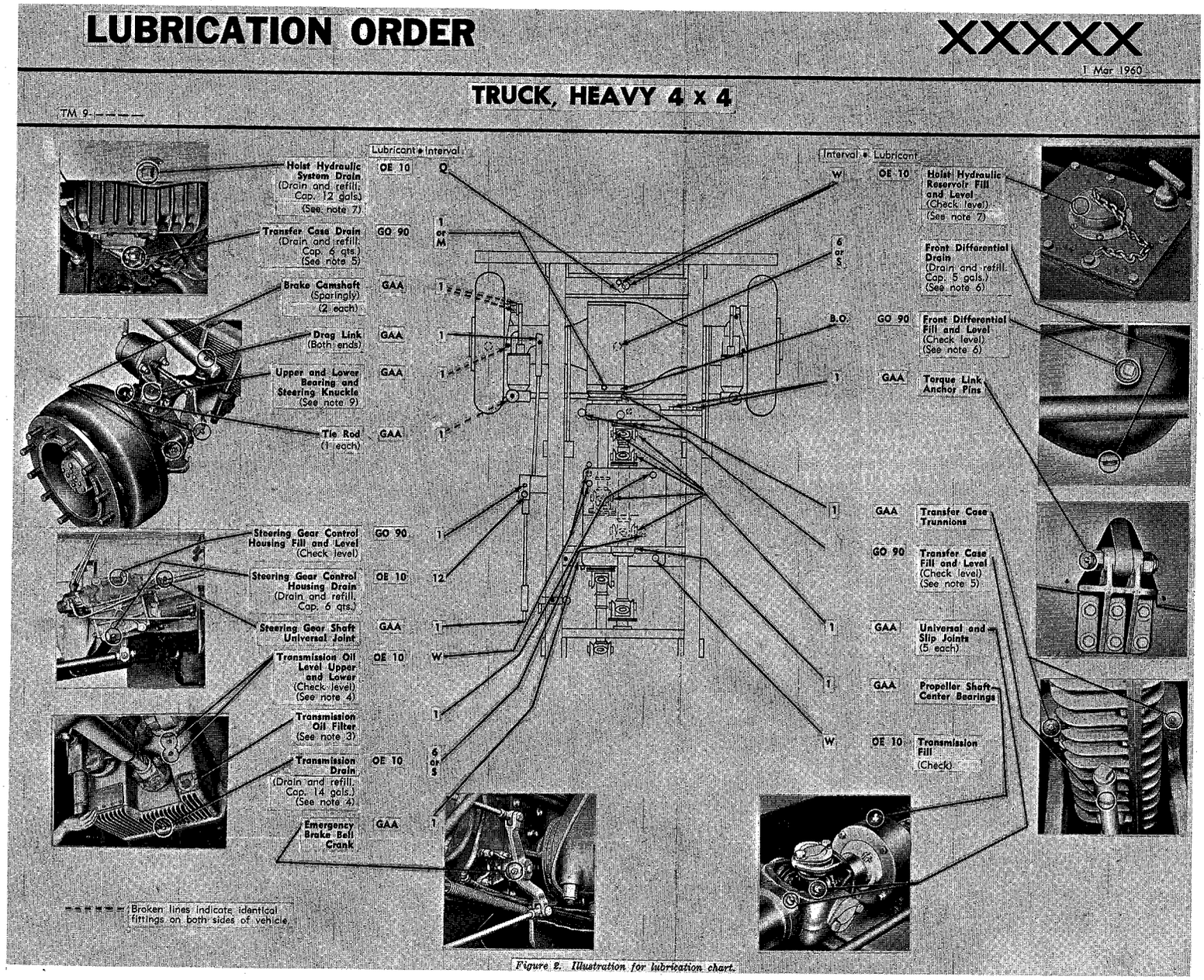


Figure 2. Illustration for lubrication chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

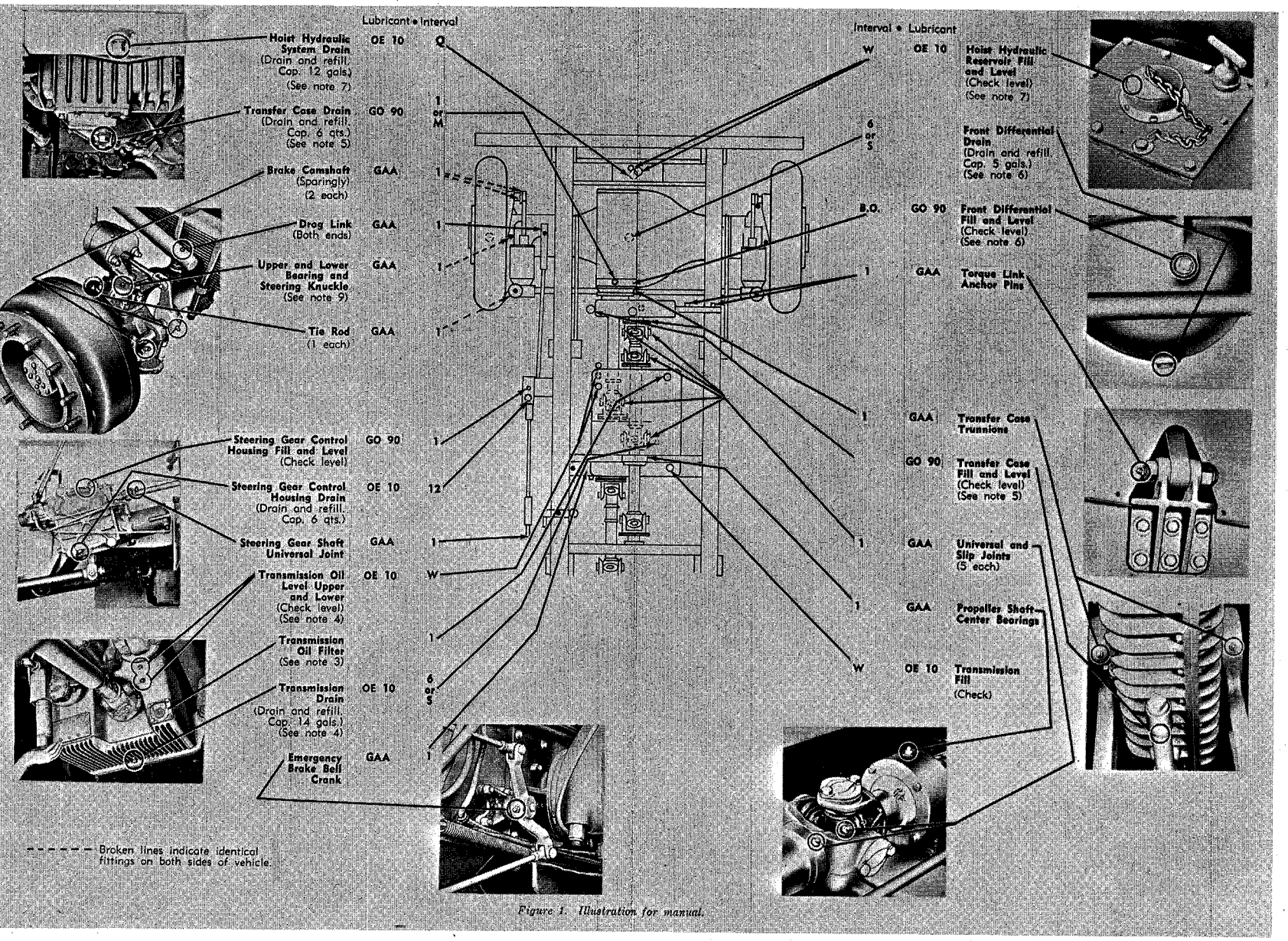


Figure 1. Illustration for manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.

KEY

LUBRICANTS SPECIFIED ON CHART ARE FOR TEMPERATURES ABOVE +32°F. FOR LOWER TEMPERATURES CHANGE GRADE AS SHOWN.

LUBRICANTS		EXPECTED TEMPERATURES			INTERVAL	
		ABOVE + 32°F	+40°F TO +10° F	0°F TO −65°F		
OE LUBRICATING OIL INTERNAL COMBUSTION ENGINE MIL-L-2104	**PL** LUBRICATING OIL, GENERAL PURPOSE PRESERVATIVE (SPECIAL) MIL-L-644	OE 50	OE 30	OES	B.O.	BEFORE OPERATION
OES LUBRICATING OIL INTERNAL COMBUSTION ENGINE SUB-ZERO MIL-L-10295	**OHA** HYDRAULIC FLUID PETROLEUM BASE, AIRCRAFT & ORD MIL-H-5606	OE 10	OE 10	OES	W	WEEKLY
GO LUBRICATING OIL GEAR UNIVERSAL MIL-L-2105	**CW** LUBRICATING OIL, CHAIN, WIRE ROPE, AND EXPOSED GEAR VV-L-751	GO 90	GO 75	GOS	M	MONTHLY
GOS LUBRICATING OIL GEAR UNIVERSAL SUBZERO MIL-L-10324	**GAA** GREASE AUTOMOTIVE AND ARTILLERY MIL-L-10924	GAA	GAA	GAA	Q	QUARTERLY
		PL	PL (SPECIAL)	PL (SPECIAL)	S	SEMI ANNUAL
					1	1,000 MILES
					6	6,000 MILES
					12	12,000 MILES

FOR ARCTIC OPERATION REFER TO-(INSERT APPROPRIATE TECHNICAL MANUAL FOR USING ACTIVITY)

1. AIR CLEANERS AND BREATHERS-(OIL BATH TYPE) DAILY, REPLENISH TO OIL LEVEL MARK WITH OE CRANKCASE GRADE. EVERY 1,000 MILES CLEAN OIL CUP AND REFILL WITH OE AS ABOVE. DISASSEMBLE, CLEAN ALL PARTS, REFILL WITH OE AS ABOVE WHENEVER CRANKCASE OIL IS CHANGED. FOR DESERT OF EXTREMELY DUSTY OPERATIONS, DISASSEMBLE, CLEAN ALL PARTS AND REFILL ONCE EVERY OPERATING DAY OR MORE FREQUENTLY IF REQUIRED. WEEKLY, REMOVE VENT SYSTEM BREATHER ELEMENT (LOCATED ON ENGINE COMPARTMENT REAR WALL OF FRONT TRUCK; ENGINE COMPARTMENT FRONT WALL OF REAR TRUCK) WASH AND REFILL WITH OE AS ABOVE.

2. ENGINE CRANKCASE AND OIL COOLER-DRAIN EVERY 1,000 MILES OR MONTHLY. DRAIN ONLY WHEN HOT AFTER OPERATION TO DRAIN ENGINE OIL COOLER, REMOVE THERMOSTATIC BYPASS VALVE AT BOTTOM OF COOLER AND DISCONNECT FLEXIBLE OIL LINE AT UPPER FRONT OF COOLER TO VENT. AFTER DRAINING, INSTALL VALVE, RECONNECT FLEXIBLE OIL LINE. REFILL CRANKCASE TO FULL MARK ON GAGE. RUN ENGINE UNTIL TEMPERATURE GAGE REGISTERS 185°F TO PERMIT ENGINE OIL TO ENTER COOLER. CHECK OIL LEVEL AND BRING UP TO FULL MARK ON GAGE. CAUTION: BE SURE ENGINE OIL-PRESSURE GAGE INDICATES OIL IS CIRCULATING.

3. OIL FILTERS-ENGINE OIL FILTER: MONTHLY, REMOVE LOCKING WIRE AND DRAIN PLUG, DRAIN SEDIMENT. EVERY 1,000 MILES OR WHENEVER CRANKCASE IS DRAINED, REMOVE ELEMENT, CLEAN, DRY AND INSTALL. TRANSMISSION OIL FILTERS: EVERY 1,000 MILES OR MONTHLY, REMOVE LOCKING WIRE AND DRAIN PLUG, DRAIN SEDIMENT. EVERY 6,000 MILES, OR SEMI-ANNUALLY WHEN TRANSMISSION IS BEING DRAINED, REMOVE FILTERS, CLEAN, DRY AND INSTALL

4. TRANSMISSION-CHECK TRANSMISSION OIL LEVEL WEEKLY BY REMOVING TRANSMISSION UPPER OIL LEVEL CHECK PLUG. IF OIL APPEARS AT HOLE, TRANSMISSION IS PROPERLY LUBRICATED. IF OIL DOES NOT APPEAR AT UPPER-LEVEL HOLE, REMOVE TRANSMISSION LOWER OIL LEVEL CHECK PLUG AND CHECK TO SEE IF OIL APPEARS AT HOLE. TRANSMISSION OIL LEVEL SHOULD BE BETWEEN UPPER AND LOWER CHECK PLUG HOLES. DRAIN EVERY 6,000 MILES OR SEMI-ANNUALLY. DRAIN ONLY WHEN HOT AFTER OPERATION. WHEN OIL HAS STOPPED DRAINING FROM TRANSMISSION, START ENGINE AND RUN AT FAST IDLE FOR APPROXIMATELY ONE MINUTE TO REMOVE OIL FROM CONVERTER. TO DRAIN COOLER, REMOVE THERMOSTATIC BYPASS VALVE AT BOTTOM FRONT OF TRANSMISSION OIL COOLER. TO VENT, DISCONNECT FLEXIBLE OIL LINE AT UPPER-REAR OF COOLER. AFTER DRAINING, INSTALL VALVE AND CONNECT FLEXIBLE OIL LINE. REMOVE AND CLEAN (SEE NOTE 3) BOTH TRANSMISSION OIL FILTERS. INSTALL FILTERS AND DRAIN PLUG AND REFILL TRANSMISSION WITH APPROXIMATELY 7 GALLONS OF OIL; START ENGINE AND RUN AT FAST IDLE UNTIL TRANSMISSION OIL TEMPERATURE REACHES 185°F; POUR 7 ADDITIONAL GALLONS OF OIL INTO TRANSMISSION. STOP ENGINE AND RECHECK OIL LEVEL AS ABOVE.

5. TRANSFER (REAR TRUCK M---ONLY)-DRAIN EVERY 1,000 MILES OR MONTHLY. DRAIN ONLY WHEN HOT AFTER OPERATION. FILL TO LEVEL MARK ON GAGE, RUN ENGINE FOR 3 MINUTES AND RECHECK LEVEL. FOR TEMPERATURES ABOVE +32°F, USE "MIN" MARK ON GAGE; FOR TEMPERATURES BELOW +32°F, USE "MAX" MARK ON GAGE. EVERY 6,000 MILES OR SEMI-ANNUALLY, REMOVE SUMP PAN AND SCREEN; CLEAN, DRY AND INSTALL.

6. DIFFERENTIALS-DRAIN EVERY 6,000 MILES OR SEMI-ANNUALLY. DRAIN ONLY WHEN HOT AFTER OPERATION. FILL TO PLUG LEVEL BEFORE OPERATION AND AFTER DRAINING.

7. HOIST-WINCH HYDRAULIC SYSTEM (FRONT TRUCK M---)-QUARTERLY DRAIN HYDRAULIC SYSTEM AND REMOVE RESERVOIR COVER TO GAIN ACCESS TO OIL FILTER LOCATED AT BOTTOM-REAR OF RESERVOIR. UNSCREW FILTER FROM BUSHING BY TURNING HANDLE ON END OF FILTER. REMOVE FILTER, CLEAN, DRY, INSTALL FILTER AND TIGHTEN HAND-TIGHT BY TURNING HANDLE ON END OF FILTER. REPLACE DAMAGED OR CLOGGED FILTERS IF NECESSARY. INSTALL RESERVOIR COVER AND FILL SYSTEM. LIFT AND LOWER HOIST CYLINDERS THREE OR FOUR TIMES TO BLEED SYSTEM AND FILL HYDRAULIC LINES WITH OE. ADD OIL UNTIL LEVEL IS UP TO MARK ON BAYONET-TYPE LEVEL CHECK GAGE.

8. HOIST HYDRAULIC SYSTEM (REAR TRUCK M---)-QUARTERLY DRAIN HYDRAULIC SYSTEM AND REMOVE RESERVOIR COVER TO GAIN ACCESS TO TWO OIL FILTERS LOCATED AT BOTTOM-RIGHT IN RESERVOIR. UNSCREW FILTERS FROM BUSHINGS BY TURNING HANDLE ON END OF EACH FILTER. REMOVE, CLEAN, DRY, INSTALL FILTERS AND TIGHTEN HAND-TIGHT BY TURNING HANDLE ON END OF EACH FILTER. REPLACE DAMAGED OR CLOGGED FILTERS IF NECESSARY. INSTALL RESERVOIR COVER AND FILL SYSTEM. LIFT AND LOWER HOIST CYLINDERS THREE OR FOUR TIMES TO BLEED SYSTEM AND FILL HYDRAULIC LINES WITH OE. ADD OIL UNTIL LEVEL IS UP TO MARK ON BAYONET-TYPE LEVEL CHECK GAGE.

FIGURE 2A. ILLUSTRATION OF NOTES AND KEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:23Z
Check the source to verify that this is the current version before use.